Sparse Hecke constraints, target connection and explicit extension traces

2026-10-30. **Finite correctness and relation-stage results; no qualified IC speedup or sub-rho claim.** The four requested studies ran, followed by target-growth and arity controls and an attempted larger conductor-30 continuation.

Current baseline update: the later pair-index reuse study harvests relations from the index already needed for target descent. It reduces the required generic queries9→1,12→1 and45→1 in three cells; the sparsest cell remains at12. The generic pair-only control closes two systems without Hecke. Read the earlier scalar-only comparison below as its exact historical method, not the strongest generic baseline. Same-total-budget superiority is unestablished.

Main result: information survives sparse projections

For D_K=-39, conductor6 and degrees2,3, all three selected fields retain64 distinct projected points. Negation folding gives32 unknown logs. At subgroup orders16,319 and206,477, doubling/tripling supplies no relations, but Hecke supplies20 independent constraints on those32 unknowns. Only12 generic base-log queries are then needed, versus32 for the frozen scalar-only bootstrap. All labels and solved target scalars were independently replayed.

Conductor / Field p / Subgroup r / Raw / unique projected / folded columns / Generic free logs / Hecke + scalar free logs / Initial target result

6 / 11,587 / 769 / 64 / 64 / 32 / 29 / 9 / verified, attempt1

6 / 163,567 / 16,319 / 64 / 64 / 32 / 32 / 12 / verified, attempt20

6 / 1,032,607 / 206,477 / 64 / 64 / 32 / 32 / 12 / no match in256 attempts

30 / 231,559 / 46,273 / 320 / 318 / 159 / 157 / 45 / verified, attempt1

This is a reduction in required generic base-label queries within the frozen method. It is not a measured reduction in complete setup or online wall time: geometric construction and linear-algebra costs also matter. The counted generic query work, including each query's scalar replay, was219 vs825;858 vs2408; 3836 vs10356; and5941 vs20526 group-addition calls respectively. Those counters exclude relation construction and LA and do not price the complete pipeline.

Exact objects and prior work

The DLP curve is always E11=X_0(11): $y^2+y=x^3-x^2-10x-20$ over Fp. D_K=-39 labels the auxiliary CM curves parameterized by modular points; it is not a claim that E11 itself has that endomorphism discriminant. Each receipt records N, r, h=N/r, the exact generator, public target, point set and full EC1/UID preimage. The four identities are registered in the canonical curve catalog; the actual projected bases are archived in archive-manifest.json. The paired crypto PR mirrors those exact curve records, without new isogeny routes.

We reviewed the original D39 result, its copied Hecke producer, the repeated-root audit, the earlier dense scalar-closure control, and the lead atlas's class-group/ECDLP bridge and representative-selection records. These prevent treating class-group relations or dense scalar collisions as an ECDLP reduction. The eta/formal-series implementation follows the existing producer and the differential normalization documented in Sage's modular-parametrization reference. FLINT0.9.0's Hilbert class polynomial routine supplies the integer CM polynomials. The attempted online FLINT reference fetch returned403; package API inspection and the exact saved Sage fixture were available. No novelty claim is made.

1. Fixed-geometry sparse fields

The primary protocol was committed and pushed before selection. The deterministic selector uses $p=v^2+39f^2/4$, the first eligible candidate at each frozen threshold, and r>=p/20. Every candidate/rejection is in its receipt. No field was selected on rank or target outcome. The public target is generated by a fixed hash-to-x/lift/projection rule without supplying its scalar.

The new finite-field implementation first reproduced all64 saved p=367 points, all48 degree2/3 rows, rank40 and128 actual rational/extension-trace fiber checks. For new fields every class-polynomial stratum has the expected number of roots and

lifts, and every auxiliary j's trace magnitude is independently counted. These are class-polynomial strata; a separate Sage endomorphism-order audit was not run. No new Sage process or unchecked installation was used.

The small, medium, large and conductor30 receipts retain points, rows, ranks, source hashes, generator/target and counters. Every attempt has a launch record, cap, exit and raw stdout/stderr. All four primary cells exited0. Fixed-seed random bases use the same point count, negation policy and multiplier budget; their exact scalar ranks are in the receipts.

2. Effective information, including new trace columns

Each system was checked on original columns, deduplicated nonzero projections and negation-folded columns. FLINT dense elimination agrees with a separate sparse modular algorithm. A separate short-curve verifier checks raw Hecke rows, projected involution rows, normalized labels and returned scalars. The independent receipt also recomputes effective ranks with a third elimination implementation.

For an extended matrix [A_old A_new], the rank of constraints that survive elimination of new columns is rank([A_old A_new])-rank(A_new). The report therefore distinguishes full rank from information on the original logs.

Trace experiment / Original folded columns / Baseline rank / Full extended columns / rank / Rank on original columns / Extra original constraints

f6, r769, degrees2,3,5,7 / 32 / 23 / 129 / 118 / 31 / 8

f30, r46273, degrees2,3,5 / 159 / 114 / 480 / 442 / 121 / 7

f6, r206477 continuation, degrees2,3,5,7 / 32 / 20 / 164 / 120 / 20 / 0

The last row is particularly discriminating:132 new columns and100 more rank do not provide a single extra constraint on the original32 logs. All three rows are finite observations, not a general scaling theorem.

3. Target connection and continuation after failure

The frozen bootstrap solves only free base logs using a shared generic BSGS table, then recovers pivot logs by LA and replays every label. This generic work is target-independent setup; it is not omitted or presented as a result of Hecke rank alone. Signed two-sum queries then try Q+[a]G for a=0..255. The small r=769 target is already represented in the projected base; its attempt1 is a coverage correctness control, not evidence of new descent. The same target is separately solved by BSGS for correctness. That reference is not matched optimized rho, so distance to rho and qualified speed are unknown.

The original large-field target failed all256 attempts. Its follow-up retains that failure and derives known labels for actual orbit traces where there is exactly one nonlinear orbit and all rational neighbors are already labeled. This needs no new generic log query. All new labels and recovered target scalars are checked by the independent short model.

Same large-field target, exploratory continuation / Known points incl. zero / Growth additions / Pair-index additions / Query attempts

Original base / 65 / 0 / 2145 / failed at256

Trace-grown / 201 / 528 / 20301 / verified at3

Generic growth, same growth-addition budget / 869 / 528 / 378015 / verified at2

Generic growth, matched201-point base / 201 / 70 / 20301 / verified at10

The growth receipt and matched-size control show an executable finite target connection after the charged bootstrap. They do not establish Hecke-specific online superiority: the same-size generic base is cheaper to grow, whereas the trace base uses fewer queries on this already observed target. Fiber-construction and verification costs are additional. This is generic decomposition with geometric preprocessing, not an algebraic PDP solver or evidence of an asymptotic sub-rho algorithm. No post-result target is called a holdout.

A final arity continuation reused the original2107-entry pair index without growing the base. The retained result found no three-term match in65 signed-base-offset attempts and no four-term match in256 sorted pair-offset attempts. Neither arm repaired this target within its cap. These are bounded negatives, not impossibility statements.

4. Degrees, conductor and actual extension traces

On the same complete conductor30 point set, degrees2,3 give folded rank100 on159 columns; adding degree5 gives rank112. Thus degree5 adds12 independent constraints before the common scalar rows. On f6, degrees5 and7 have no complete rational fibers in the checked field, but their extension fibers are computed rather than represented by unknown formal symbols.

The neighbor y-coordinate is recovered from the trace-moment matrix and verified against the curve equation in the quotient ring. Each irreducible orbit is summed under Frobenius using explicit extension-field arithmetic. Its rational trace point is checked, as is the sum-of-orbits Hecke identity. The original and continuation f6 fields have patterns[1,1,4] for degree5 and[4,4] for degree7. No repeated-x fiber occurred in these cells; repeated-x inputs remain an explicit unsupported status, not an assumed successful row.

The larger conductor30 continuation did not select an eligible field in its frozen200-candidate window. The retained selector record and exit1 launch show missing evidence, not a failed mathematical claim. Its threshold or subgroup condition was not silently relaxed.

Validation, limits and next work

Four arithmetic/information-accounting unit tests pass, including an independently calculated quadratic-orbit trace. All primary receipts and all new labels have independent short-model group replay. The initial signed-eigenvalue bug, its source snapshot and failed control are retained under runs/control-01; the corrected full control passes. The four factor-base archives rebuild byte-for-byte from source-bound receipts. Curve identity metadata validates locally and matches the paired crypto mirror.

This host is CPU-only; hardware and execution record observable placement/limits. Physical memory technology is unknown. Wall times are diagnostic aggregate times, not isolated comparisons. Exclusive phase timers and a matched strong-rho run remain missing, so complete costs and speedups are unknown. These runs do not ratchet the best verified full IC baseline. The narrow observed advance is fewer generic base-label queries.

Proof map: finite curve equations, group rows, ranks and scalar outputs are checked by the retained code/certificates. General identification of all formal-series functions with Hecke correspondences relies on the sourced construction; a kernel-checked general theorem and Lean versions of the finite certificates remain open. Lean availability records that Lean4/mathlib are absent. No weaker theorem or unbuilt source is claimed as a checked proof.

Requested step / Execution status / Remaining goal

1: fixed geometry, sparse subgroups / verified finite panel / larger sizes and held-out scaling

2: effective constraints / verified, including elimination of new trace columns / structural/formal explanation

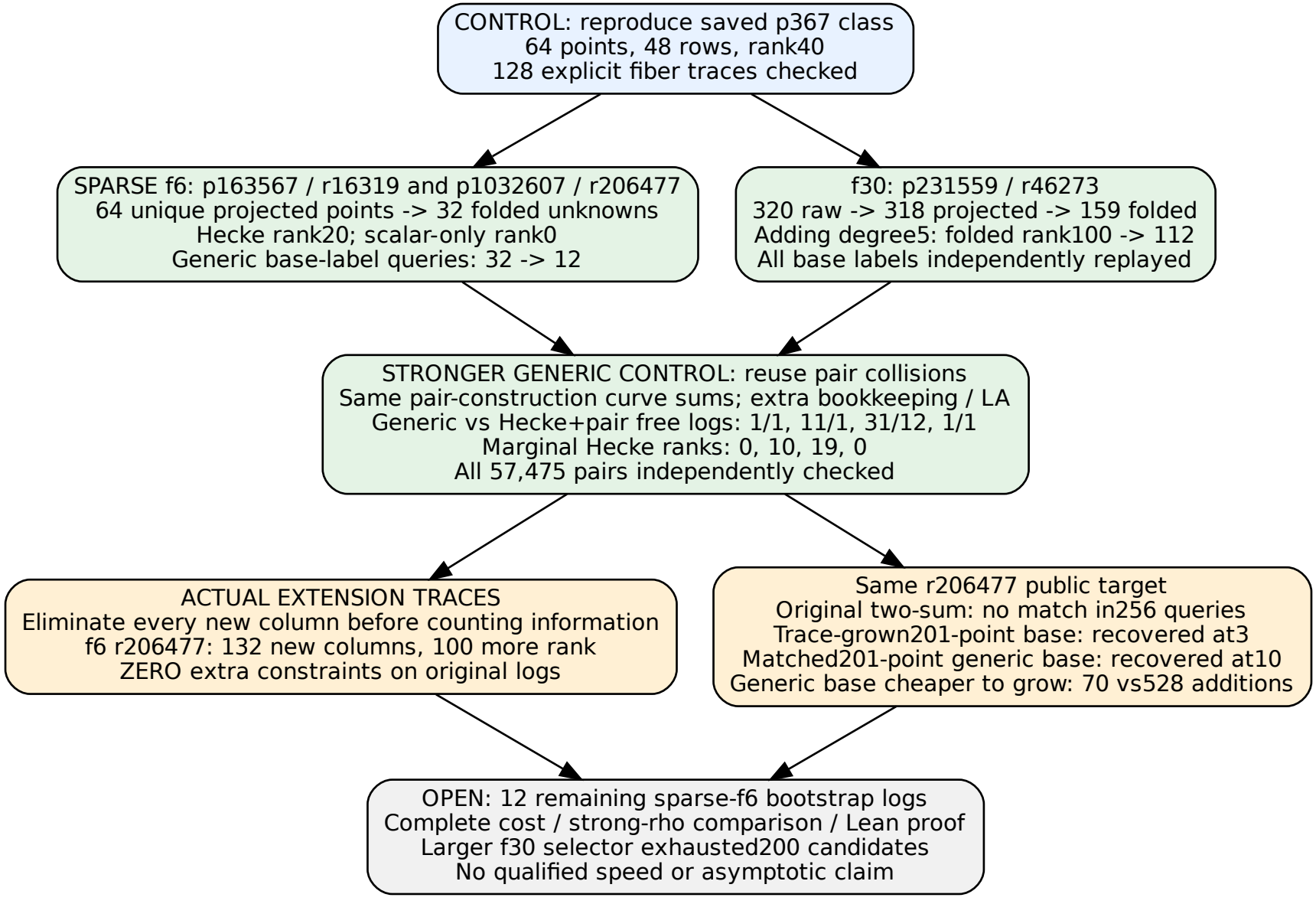
3: target connection / verified bounded implementation; one initial failure, then verified growth recovery / cheaper scalable descent and matched rho

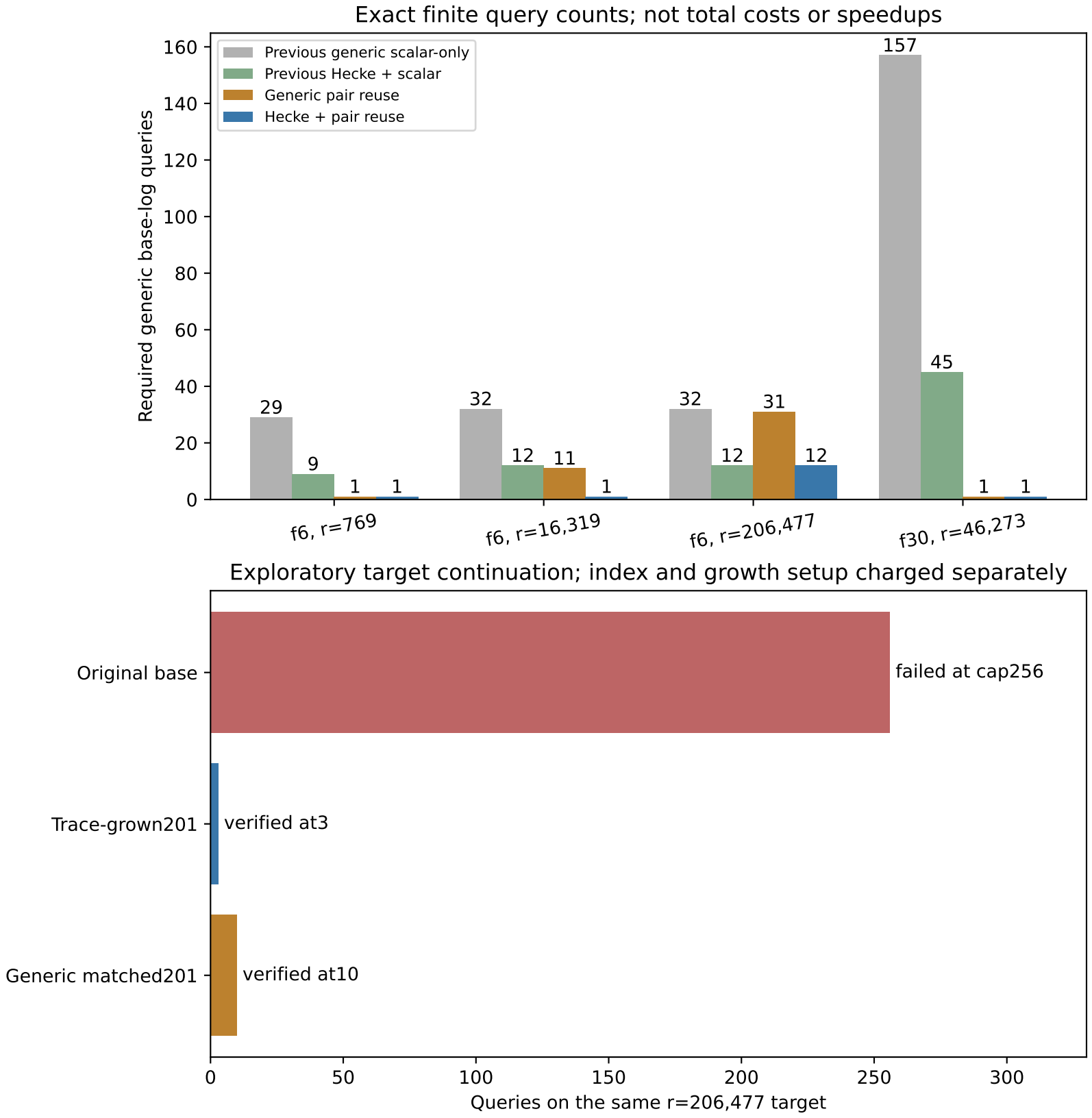
4: richer Hecke relations / verified degrees/conductor/extension cells / larger f30 continuation missing within its selector cap

PR and continued pursuit / reports, source, receipts, archives and paired metadata PR / hosted gates; next registered experiment

Next: reduce the remaining12 generic bootstrap queries in sparse f6 without paying for unanchored trace columns, or construct a cheaper known-label target oracle. The measured candidate should then face fresh single-target controls, complete phase accounting and matched strong rho on the required benchmark host. The existing problem record tracks these exact gaps; recording this result does not resolve the full goal.

The diagram, plots and PDF are rebuilt from editable sources. Existing dense-D39 and surface-transfer diagrams were checked; this changes relation evidence, not isogeny-route edges, so those canonical graphs stay unchanged. The lead-atlas review should link this finite bootstrap/target result rather than treating it as a class-group/ECDLP reduction.


CONTROL: reproduce saved p367 class
64 points, 48 rows, rank40
128 explicit fiber traces checked
SPARSE f6: p163567 / r16319 and p1032607 / r206477
64 unique projected points -> 32 folded unknowns
Hecke rank20; scalar-only rank0
Generic base-label queries: 32 -> 12
f30: p231559 / r46273
320 raw -> 318 projected -> 159 folded
Adding degree5: folded rank100 -> 112
All base labels independently replayed
STRONGER GENERIC CONTROL: reuse pair collisions
Same pair-construction curve sums; extra bookkeeping / LA
Generic vs Hecke+pair free logs: 1/1, 11/1, 31/12, 1/1
Marginal Hecke ranks: 0, 10, 19, 0
All 57,475 pairs independently checked
ACTUAL EXTENSION TRACES
Eliminate every new column before counting information
f6 r206477: 132 new columns, 100 more rank
ZERO extra constraints on original logs
Same r206477 public target
Original two-sum: no match in256 queries
Trace-grown201-point base: recovered at3
Matched201-point generic base: recovered at10
Generic base cheaper to grow: 70 vs528 additions
OPEN: 12 remaining sparse-f6 bootstrap logs
Complete cost / strong-rho comparison / Lean proof
Larger f30 selector exhausted200 candidates
No qualified speed or asymptotic claim


Exact finite query counts; not total costs or speedups
Previous generic scalar-only
Previous Hecke + scalar
Generic pair reuse
Hecke + pair reuse
Required generic base-log queries
0
20
40
60
80
100
120
140
160
29
9
1
1
32
12
11
1
32
12
31
12
157
45
1
1
f6, r=769
f6, r=16,319
f6, r=206,477
f30, r=46,273
Exploratory target continuation; index and growth setup charged separately
Original base
failed at cap256
Trace-grown201
verified at3
Generic matched201
verified at10
0
50
100
150
200
250
300
Queries on the same r=206,477 target